

**Investigation of How The Coefficient of Restitution of a Tennis
Ball–Racket System Varies With Impact Velocity
Physics Standard Level May 2026**

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INTRODUCTION

Tennis has been a big passion of mine since I was young. Being on the varsity tennis team in high school exposed me to competition regulations. I noticed how each match was played with the same set of balls and opened new to ensure fair play. During the tournament, the balls felt bouncier than mine from home. As I played, I had to adjust to the feel of the competition balls to ensure they did not go out of the court.

I decided to measure the “bounciness” of the ball by finding the coefficient of restitution (C_R). Although we have not learned about the coefficient of restitution in class, I researched that it is a measure of the fraction of kinetic energy that objects possess after colliding with one another in a ratio compared to before they collide. With values ranging between 0 and 1, coefficient of restitution varies from ball to ball and depends on the ball’s makeup, including factors such as internal pressures and resilience of material. According to Andre Roux, a tennis ball has a coefficient of restitution around 0.728 - 0.759. C_R is found by the equation where v_f is the speed of the ball right after impact and v_i is the speed of the ball right before impact.

$$C_R = \frac{v_f}{v_i}$$

RESEARCH QUESTION

How does the impact velocity of a tennis ball, ms^{-1} , affect the coefficient of restitution, C_R , when dropped from varying heights (200 cm, 175 cm, 150 cm, 125 cm, 100 cm, 75 cm, 50 cm, 25 cm) onto the rim-side of a tennis racket?

THEORETICAL RELATIONSHIP AND HYPOTHESIS

When a tennis ball collides with a stationary racket, the collision can be treated as a vertical collision in which energy is not conserved. During the impact, a portion of the ball’s kinetic energy is stored as elastic potential energy and a small portion is dissipated as thermal energy or sound. The extent of this energy loss is characterized by the coefficient of restitution (C_R).

The relationship between impact velocity and drop height follows that the ball is released from rest and falls vertically under the influence of gravity on earth, 9.8ms^{-2} . Assuming air resistance is negligible over the short fall distance, the impact velocity of the ball can be calculated using conservation of energy equations. When a ball is at a certain height above a surface, potential energy is at a maximum. But, as it is dropped and right before it makes impact, that potential energy is converted into kinetic energy. The moment the ball touches the floor, a fraction of that kinetic energy is lost through sound and heat. This is assumed that energy is conserved before impact, meaning that the potential energy (E_p) will be equal to the kinetic

energy (E_k) of the ball just before it makes contact with the surface. Seen below, g is the acceleration due to gravity and h_1 is the vertical drop height.

$$E_p = E_k$$

$$mgh = \frac{1}{2}mv_i^2$$

$$v_i = \sqrt{2gh_1}$$

This relationship shows that impact velocity (v_i) increases with increasing drop height. Thus, the experiment will allow impact velocity to be controlled experimentally by varying the release height of the ball. After the collision, the ball rebounds upward and reaches a maximum height determined by its rebound velocity. By rearranging the previous kinematic equation:

$$h = \frac{v_i^2}{2g}$$

The coefficient of restitution can be expressed in terms of height:

$$C_R = \frac{\sqrt{2gh_2}}{\sqrt{2gh_1}}$$

$$C_R = \sqrt{\frac{h_2}{h_1}}$$

Where h_1 is the height of the ball before a bounce and h_2 is the height after the bounce. This equation allows the coefficient of restitution to be calculated directly from experimentally measured bounce heights, eliminating the need to measure velocities explicitly.

A coefficient of restitution of 0 represents a perfectly inelastic collision, in which the ball would not rebound, while a coefficient of 1 represents a perfectly elastic collision, in which no kinetic energy is lost and the rebound velocity equals the impact velocity. As the impact velocity of a tennis ball on a tennis racket increases, a greater proportion of energy is dissipated through sound, heat, and deformation of the ball. Thus, it is hypothesized that as the impact velocity of the tennis ball increases, the coefficient of restitution (C_R) decreases.

VARIABLES

Variable and Type	Description
Independent: The drop height from	A tape measurer is taped parallel to the freefall of the tennis ball onto a wall that is perpendicular to the floor where the tennis racket lies. Range and Increment: The range of the drop height will be in increments of 25 cm for

which the tennis ball will fall from	eight trials: 200 cm, 175 cm, 150 cm, 125 cm, 100 cm, 75 cm, 50 cm, 25 cm. Uncertainty: The uncertainty will be approximately ± 2.00 cm (Δh_1). The bottom of the ball will be placed directly on the tape measurer's line for its respective drop height, held by hand, resulting in slight human error. The uncertainty is accounted for and will be combatted against by using several trials. Why: The selected drop heights (25 cm to 200 cm) were chosen to provide a sufficient range of gravitational potential energy to observe measurable differences in the tennis ball's coefficient of restitution. Increments of 25 cm will be used as they are large enough to produce noticeable changes in the dependent variable.
Dependent: Coefficient of Restitution	The coefficient of restitution will be calculated from the drop and bounce heights obtained from the Vernier Motion Sensor and LabQuest 3 Uncertainty: The Vernier motion sensor displays displacement values to three decimal places in metres. Since it is a digital device, the uncertainty should be ± 1 the least significant digit. Therefore, the absolute uncertainty in position measurements is ± 0.001 m, equivalent to ± 0.1 cm.
Controlled	Description
Tennis ball (pressure, material)	How: Using the same new Wilson Trinity 1 ball, with a ~25-27 psi listed online Why: Different tennis balls have different internal pressures, elasticity, and material which affect "bounciness" (C_R) of the ball.
Tennis racket (string tension)	How: Using the same Head Challenger tennis racket Why: Different rackets have different string tensions and frame stiffness, which influence energy transfer during collision. The strings lose tension and elasticity over time.
Racket position (impact location)	How: Racket is not moved during the experiment as it sat on the floor. To ensure it did not move, a textbook is placed on top of it. Why: The stiffness of the racket varies across the string bed.
Release method	How: The ball falls from hand by releasing hold of it. No force is exerted, as it is a freefall with only gravity acting on it. Why: Adding any initial velocity with force would change the impact velocity.
Surface and environment	How: Experiment conducted in the basement of a house at room temperature (~20 degrees celsius), on laminate flooring. Why: Temperature affects the elasticity of the ball and racket strings. Contact with different surfaces could introduce secondary collisions.
Motion sensor placement	How: The Vernier motion sensor is fixed directly above the drop height and impact point, and is not moved. Why: As the motion sensor picks up on sounds, changing the sensor position would affect distance and velocity measurements, introducing systematic error.

MATERIALS

1. Tennis ball (Wilson Trinity 1)
2. Tennis racket (Head Challenger)
3. Measuring tape with centimetre scale (absolute uncertainty: ± 0.05 cm; percent uncertainty: $\approx \pm 0.025\%$ at 200 cm)
4. Adhesive tape
5. Vernier LabQuest 3
6. Vernier Motion sensor (absolute uncertainty: ± 0.1 cm; percent uncertainty: $\approx \pm 0.05\%$ at 200 cm)
7. Clamp to keep the sensor in place
8. Chair to reach the drop height

PROCEDURE

The measuring tape is secured vertically to a wall using adhesive tape, ensuring it is perpendicular (90°) to the floor. The tennis racket is placed flat on the floor directly below the measuring tape, centered beneath it. The motion sensor is clamped onto a wall hook high enough to take measurements for all drop heights. The LabQuest is connected to the motion sensor with a wire to collect the data. For the first drop height, position the bottom of the tennis ball directly on the corresponding line of the measuring tape, very close to, but not touching the wall. Press the green play arrow on the LabQuest to begin the detection. Release the ball without applying any additional force and stop the motion sensor recording after the ball has made impact with the racket. Repeat this trial 5 times. Then, repeat the entire process for the remaining drop heights. Save and collect all data through the Vernier LabQuest 3 detector.

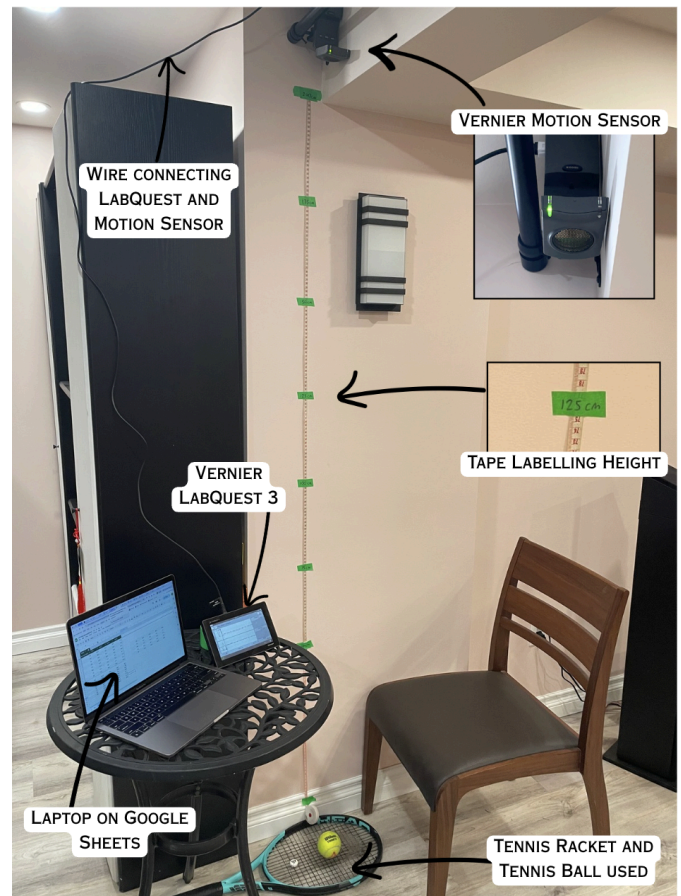


Fig 1. Labeled Photo of Experimental Setup

DATA COLLECTION

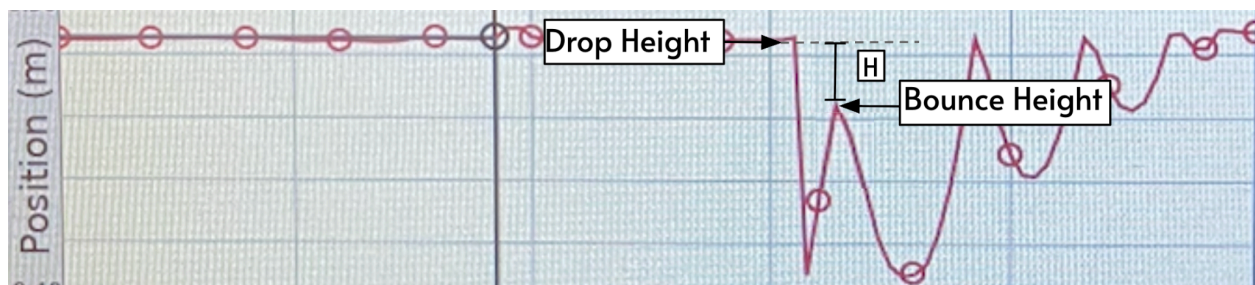


Fig 3. An example of a position vs. time graph the Vernier LabQuest produces from the tennis ball bounce. The

trough shows the position change taking place at the first bounce. The difference between the drop and bounce heights is labelled H

Reading the Vernier LabQuest 3 Graph display:

Upon beginning a time-based data collection, continue until the ball has reached its maximum height after one bounce at least. To determine the dependent variable values, interpret the sudden change (displayed in the graph as the trough) in the position vs. time graph produced by the motion detector (Fig. 3). The initial stable, horizontal line position represents the drop height and the position at the end of trough is the bounce height. The difference, H, between the two heights will provide the maximum rebound height reached after one bounce.

METHOD JUSTIFICATION

Dropping the tennis ball onto the rim-side:

When initially experimenting to have the tennis ball fall onto the centre of the racket, the impact caused the strings to deform significantly which transferred a large impulse to the racket, and striking the floor. To minimize unwanted energy losses, the procedure was modified so that the tennis ball was released closer to the wall and allowed to strike the side of the racket instead of the center. Since tennis racket strings are looser in the middle and tighter at the sides, this configuration prevented the strings from contacting the floor.

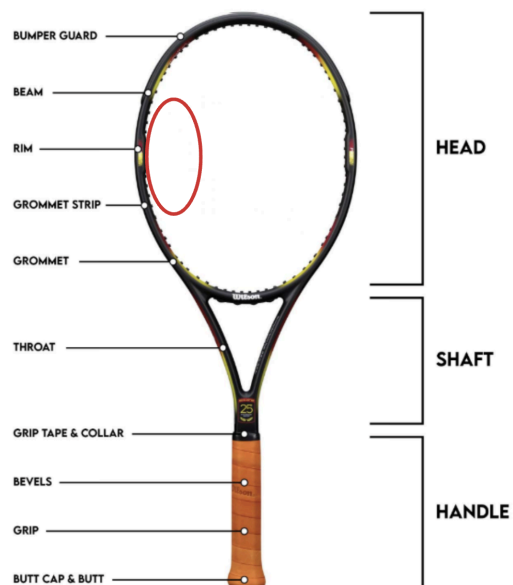


Fig 2. Location of tennis ball impact spot

RISK ASSESSMENT AND SAFETY

- Safety considerations: work in a spacious environment in order to avoid accidents caused by stray tennis balls. The racket is stored away once the experiment is complete to avoid tripping.
- Environmental considerations: Discard waste (tape) into the correct categories and turn off the LabQuest to preserve energy.
- Ethical considerations: not applicable

RAW DATA

Drop Height ($h_1 \pm 2$ cm)	Maximum Height Reached After One Bounce ($h_2 \pm 0.1$ cm)				
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
200	97.6	102.4	104.9	106.8	104.2
175	93.6	96.5	99.4	94.7	90.8

150	84.5	92.9	88.6	91.8	88.0
125	64.3	67.4	70.9	73.8	69.4
100	56.7	60.5	58.8	57.9	57.8
75	42.1	41.0	43.9	43.8	43.7
50	30.1	29.0	31.8	32.2	31.4
25	15.3	15.2	17.4	16.9	17.1

PROCESSED DATA

The average rebound height is calculated to obtain a reliable value for determining the coefficient of restitution. Finding the mean of the five trials requires adding each trial and dividing that sum by 5. Using the 200 cm drop as an example:

$$h_{\text{avg}} = \frac{h_{2,1} + h_{2,2} + h_{2,3} + h_{2,4} + h_{2,5}}{5}$$

$$h_{\text{avg}} = \frac{97.6 + 102.4 + 104.9 + 106.8 + 104.2}{5}$$

$$h_{\text{avg}} = 103.2 \text{ cm}$$

The random uncertainty is taken as the standard deviation of the trials:

$$\sigma = \sqrt{\frac{\sum (h_z - h_{\text{avg}})^2}{n-1}}$$

In the equation above, sigma (standard deviation) is equal to the square root of the sum of the square of each trial's deviation from the mean value of the set ($h_{\text{avg}} = 103.2 \text{ cm}$) divided by 4 (number of trials subtract one = $5 - 1$)

Finding the deviation from the mean:

$$\begin{aligned} 97.6 - 103.2 &= -5.6 \\ 102.4 - 103.2 &= -0.8 \\ 104.9 - 103.2 &= 1.7 \\ 106.8 - 103.2 &= 3.6 \\ 104.2 - 103.2 &= 1.0 \end{aligned}$$

Below, the numerator displays the sum of the squares of the 5 standard deviations listed above:

$$\sigma = \sqrt{\frac{\sum (h_z - h_{\text{avg}})^2}{n-1}}$$

$$\sigma = \sqrt{\frac{31.36 + 0.64 + 2.89 + 12.96 + 1.00}{5 - 1}}$$

$$\sigma = \sqrt{\frac{48.85}{4}}$$

$$\sigma = \sqrt{12.21} = 3.49 \text{ cm}$$

The sample standard deviation formula is applicable because the same measuring instrument is used each time and variations come mainly from random effects (using a hand to drop and impact angle). This calculation of uncertainty is more favourable over instrument uncertainty because the LabQuest's uncertainty is much smaller than the spread of the data. So, the standard deviation is used as the uncertainty for h_{avg} :

$$h_{\text{avg}} \pm \sigma \text{ cm}$$

Drop Height ($h_1 \pm 2 \text{ cm}$)	Mean Bounce Height, h_{avg} (cm)	Uncertainty, σ (cm)
200	103.39	± 3.49
175	95.05	± 3.21
150	88.01	± 3.33
125	69.19	± 3.58
100	58.38	± 1.42
75	42.75	± 1.29
50	30.73	± 1.32
25	16.64	± 1.05

Calculating the impact velocity for the 200 cm drop:

Firstly, converting the drop height to meters:

$$h_1 = 2.00 \pm 0.02 \text{ m}$$

$$v_i = \sqrt{2gh_1}$$

$$v_i = \sqrt{2(9.81 \text{ ms}^{-2})(2.00 \text{ m})}$$

$$v_i = 6.26 \text{ ms}^{-1}$$

Finding the uncertainty for the impact velocity is derived from the identification that 2 and g are constants and do not contribute to uncertainty, while only height (h_1) has uncertainty. Applying the power rule since $v \propto \sqrt{h_1}$, the fractional uncertainty is given by:

$$\frac{\Delta v_i}{v_i} = \left(\frac{1}{2}\right)\left(\frac{\Delta h_1}{h_1}\right)$$

$$\Delta v_i = v_i \left(\frac{1}{2}\right)\left(\frac{\Delta h_1}{h_1}\right)$$

$$\Delta v_i = 6.26 \left(\frac{1}{2}\right)\left(\frac{0.02}{2.00}\right)$$

$$\Delta v_i = 0.03 \text{ ms}^{-1}$$

Thus, the impact velocity of the tennis ball from a drop height of 200 cm is:

$$v_i = 6.26 \pm 0.03 \text{ ms}^{-1}$$

Calculating the coefficient of restitution for the 200 cm drop:

$$C_R = \sqrt{\frac{h_{avg}}{h_1}} = \sqrt{\frac{103.39}{200}} = 0.719$$

Deriving the uncertainty for the coefficient of restitution follows the principals that multiplication/division involves adding fractional uncertainties, and powers involve multiplying by the absolute value of the power. Uncertainties are propagated using the quadrature method, as the uncertainties in drop height and rebound height are independent and random. The general formula for finding the uncertainty for C_R is as follows:

$$\frac{\Delta C_R}{C_R} = \left(\frac{1}{2}\right) \sqrt{\left(\frac{\sigma}{h_{avg}}\right)^2 + \left(\frac{\Delta h_1}{h_1}\right)^2}$$

$$\Delta C_R = C_R \left(\frac{1}{2}\right) \sqrt{\left(\frac{\sigma}{h_{avg}}\right)^2 + \left(\frac{\Delta h_1}{h_1}\right)^2}$$

Solving for the coefficient of restitution uncertainty of the 200 cm drop:

$$\Delta C_R = 0.719 \left(\frac{1}{2}\right) \sqrt{\left(\frac{3.49}{103.39}\right)^2 + \left(\frac{2}{200}\right)^2}$$

$$\Delta C_R = 0.719\left(\frac{1}{2}\right) \sqrt{(0.034)^2 + (0.010)^2}$$

$$\Delta C_R = 0.013$$

The coefficient of restitution is reported to three decimal places in the processed data section in order to avoid premature rounding during calculations. This precision is favourable for analysis and graphical representation. The final coefficient of restitution for the drop at 200 cm:

$$C_R = 0.719 \pm 0.013$$

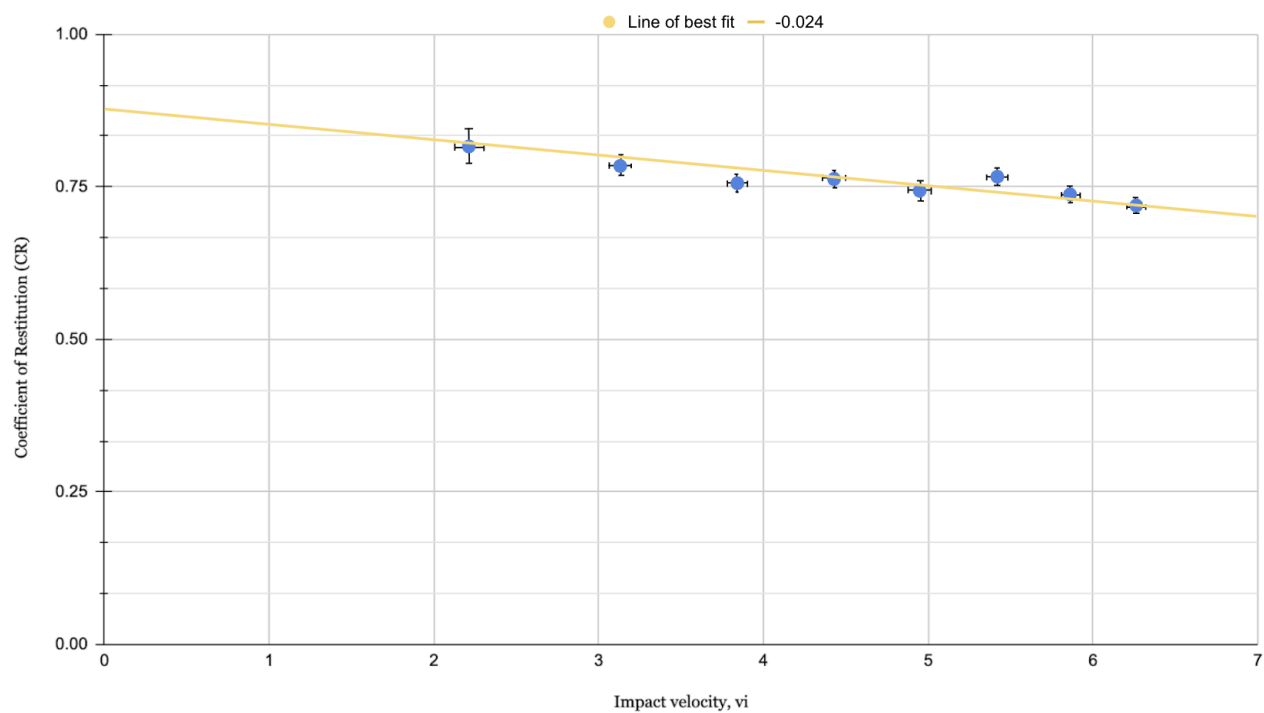
Since the coefficient of restitution is a dimensionless quantity, it has no units. The following table displays the impact velocity and uncertainty, and coefficient of restitution and uncertainty for all the experimented drop heights.

Drop Height ($h_i \pm 2$ cm)	Impact velocity, v_i (ms^{-1})	Uncertainty, Δv_i (ms^{-1})	Coefficient of Restitution (C_R)	Uncertainty, ΔC_R
200	6.26	± 0.03	0.719	± 0.013
175	5.86	± 0.03	0.737	± 0.013
150	5.42	± 0.04	0.766	± 0.015
125	4.95	± 0.04	0.744	± 0.020
100	4.43	± 0.04	0.764	± 0.012
75	3.84	± 0.05	0.755	± 0.015
50	3.13	± 0.06	0.784	± 0.023
25	2.21	± 0.09	0.816	± 0.042

GRAPHICAL PRESENTATION

A scatter plot of coefficient of restitution, C_R against impact velocity, v_i is constructed to investigate the relationship between these two variables. Impact velocity is plotted on the x-axis as it is derived from the independent variable (drop height), while C_R is plotted on the y-axis as the dependent variable calculated from experimental measurements. By initially plotting the coefficient of restitution (C_R) against impact velocity, v_i , the relationship appears approximately linear, exhibiting a slight decrease in C_R as impact velocity increased. To quantify this trend, a linear model is fitted to the data using a line of best fit constrained by maximum and minimum gradient bounds derived from experimental uncertainty.

Coefficient of Restitution (COR) vs. Impact velocity, v_i (ms^{-1})



Coefficient of Restitution (COR) vs. Impact velocity, v_i (ms^{-1})

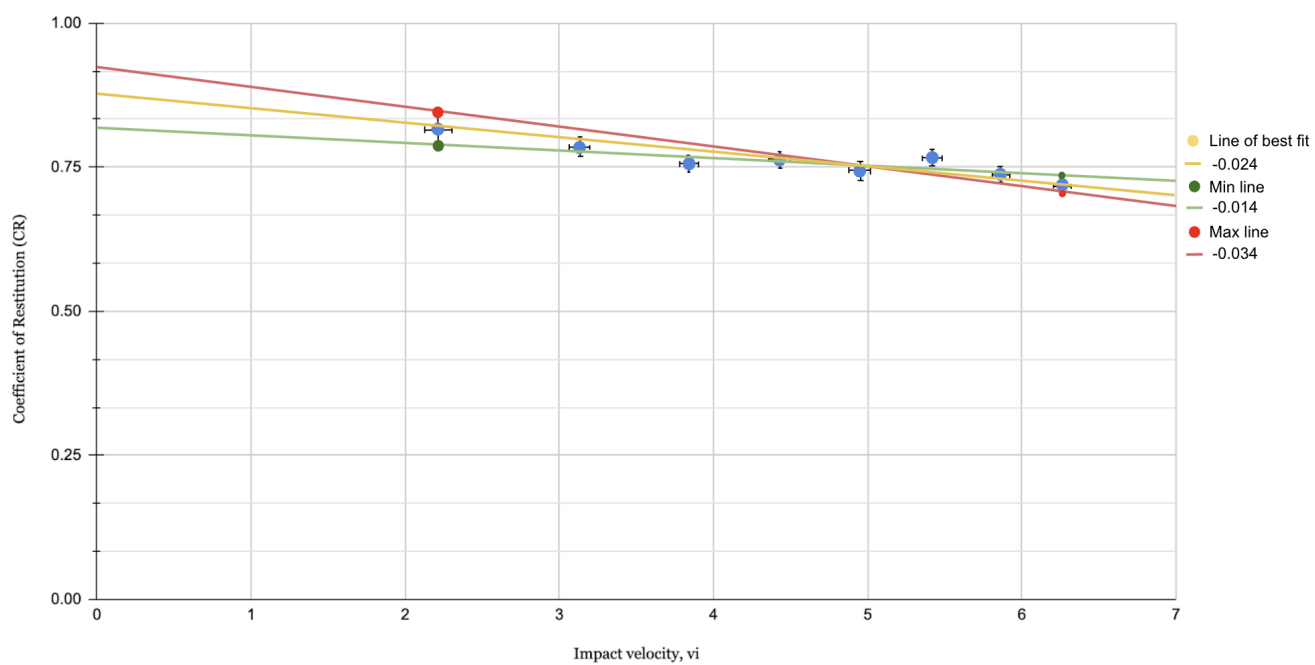


Fig. 4. A graph of C_R vs. impact velocity (ms^{-1}). Error bars determined using ΔC_R values

Gradient	Gradient uncertainty	Y-intercept	Y-intercept uncertainty
$m_{min} = -0.014 \text{ sm}^{-1}$	$\Delta m = \frac{m_{max} - m_{min}}{2}$	$Y_{int_{max}} = 0.917$	$\Delta Y = \frac{Y_{int_{max}} - Y_{int_{min}}}{2}$
$m_{max} = -0.034 \text{ sm}^{-1}$	$\Delta m = \frac{(-0.034) - (-0.014)}{2}$	$Y_{int_{min}} = 0.813$	$\Delta Y = \frac{0.917 - (0.813)}{2}$
$m = \frac{m_{max} + m_{min}}{2}$	$\Delta m = -0.01$	$Y_{int} = \frac{Y_{int_{max}} + Y_{int_{min}}}{2}$	$\Delta Y = 0.052$
$m = \frac{(-0.034) + (-0.014)}{2}$		$Y_{int} = \frac{0.917 + 0.813}{2}$	
$m = -0.024 \text{ sm}^{-1}$		$Y_{int} = 0.865$	

Therefore, the line of best fit's slope can be modelled:

$$m = -0.024 \pm 0.01 \text{ sm}^{-1}$$

Given that the y-intercept is 0.865 ± 0.052 , the line of best fit equation, written in the format $y = b + mx$, is represented by:

$$C_R = (0.865 \pm 0.052) - (0.024 \pm 0.010)v_i$$

ANALYSIS

By observing the graph of coefficient of restitution (C_R) against impact velocity (Fig. 4), it is observed that the coefficient of restitution shows a slight decreasing trend as impact velocity increases. Although individual data points fluctuate within their experimental uncertainties, the overall trend suggests a weak negative correlation between the two variables. A straight-line model is fitted to the data, yielding a gradient of:

$$m = -0.024 \pm 0.01 \text{ sm}^{-1}$$

and a line of best fit of:

$$C_R = (0.865 \pm 0.052) - (0.024 \pm 0.010)v_i$$

The uncertainty in the gradient is determined using the maximum and minimum gradient method. The best-fit gradient lies approximately midway between these two extreme values, indicating that the linear model is consistent with the data within uncertainty. However, the data points at lower impact velocities (approximately $2\text{--}3 \text{ ms}^{-1}$) have slightly larger uncertainties in C_R while at higher impact velocities (above 4 m s^{-1}), the C_R uncertainty values are closer.

Physically, at lower impact velocities, the tennis ball and racket deform more elastically, resulting in smaller energy losses and higher values of C_R . Since the y-intercept does not approach 1, this indicates that even at very low impact velocities the collision is not perfectly

elastic, which is expected due to unavoidable internal energy losses within the ball–racket system.

The largest deviation from the line of best fit occurs at an impact velocity of 2.21 m s^{-1} where C_R is 0.816 ± 0.042 . This point has the largest relative uncertainty due to the small rebound height, making percentage uncertainty larger and increasing the influence of random variations in bounce behavior. This deviation is indicative of random error rather than a contradiction to the linear model. Similarly, the data point with a C_R of 0.766 ± 0.015 lies slightly above the best-fit line but remains within uncertainty and does not significantly contradict the overall trend.

CONCLUSION

This investigation aimed to determine how the impact velocity of a tennis ball affects the coefficient of restitution when dropped onto the rim-side of a tennis racket. The results show that the coefficient of restitution decreases with increasing impact velocity, indicating a weak negative correlation between the two variables.

STRENGTHS

- The apparatus of the Vernier motion sensor and LabQuest 3 was available to use, being in abundance in high school physics storage rooms. It allowed rebound heights to be measured with high precision and minimized human reaction time and parallax error. The practicality of this tool makes the experimental procedure easy to replicate, set up, and restart.
- The use of repeated trials (five per height) reduced random uncertainty. This strengthened the validity of the mean rebound heights and coefficients of restitution.
- The experiment maintained strong control of variables, including the use of the same tennis ball, racket, and environment. This ensured that changes in C_R could be attributed primarily to changes in impact velocity rather than extraneous factors.

LIMITATIONS AND IMPROVEMENTS

Error Category	Error	Description	Effect on Results	Improvements
Human Error	Inconsistent ball release	The tennis ball was released by hand, which could introduce small variations in release position or initial rotation	Variations in impact angle and spin increased random uncertainty in rebound height.	Use a fixed release tool or clamp.

Human Error	Impact location	Although the ball was aimed at the rim-side of the racket, slight variations in where the ball struck the racket occurred between trials.	Different impact locations can alter stiffness of the racket, causing variation in rebound height.	Mark a fixed impact point on the racket frame and guide the ball along a vertical tube to ensure consistent contact.
Random Error	Erratic bouncing of the tennis ball	The ball occasionally bounced slightly sideways or rotated unpredictably after impact due to uneven contact.	Increased scatter in rebound height measurements, especially at lower drop heights where uncertainties were proportionally larger.	Increase the number of trials per drop height to reduce the influence of random scatter through averaging.
Systematic Error	Energy loss to racket motion	Although a textbook was placed on the racket, it does not perfectly represent an isolated collision as movements of the racket frame with the floor may still have occurred.	Energy was lost through vibrations transmitted into the ground, slightly lowering measured C_R values.	Clamp the racket securely to a rigid support, above ground.

EXTENSION

This investigation can be extended in many meaningful ways, still focusing on coefficient of restitution:

- Investigating how the coefficient of restitution varies with ball pressure by comparing new and used tennis balls
- Exploring the effect of racket properties, such as string tension or frame material, on the coefficient of restitution (a racket with newly strung vs. a racket that had not been strung in over 10 years)

COMPARISON WITH PUBLISHED LITERATURE

A recent study by Muangsiri in 2021 investigated the relationship between impact velocity and coefficient of restitution for table tennis balls over a wide range of velocities using a high-speed camera. The study found that the coefficient of restitution decreased linearly with increasing impact velocity, with a negative slope of approximately -0.018 per 1 m s^{-1} .

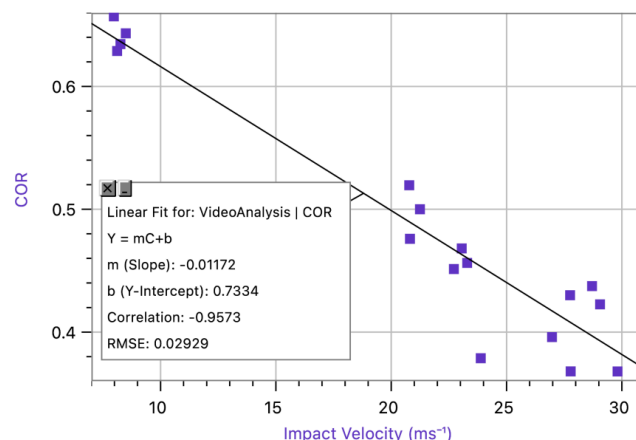


Fig 5. Kawazoe & Suzuki's results of the effect of the impact velocity of a table tennis ball on its coefficient of restitution

While the magnitude of the gradient obtained by Kawazoe and Suzuki is larger than that obtained in this experiment, this discrepancy can be attributed to differences in experimental conditions. In particular, the present investigation is limited to much lower impact velocities and a different ball-surface system (tennis ball and racket). Furthermore, Kawazoe and Suzuki note that a linear model provides a strong empirical fit within the tested velocity range, even though theoretical models may predict non-linear behavior at very low velocities.

Despite these differences, this behaviour is consistent with the trend observed in the present investigation of a tennis ball-racket system. In both studies, the coefficient of restitution decreases as impact velocity increases, indicating that greater energy dissipation occurs at higher impact speeds.

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